

Literature resolution of the cyclic-group Bohnenblust–Hille subexponentiality question

Status packet for arXiv:2310.07926

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Source question

In *Dimension-free discretizations of the uniform norm by small product sets*, Becker–Klein–Słote–Volberg–Zhang ask (Question 2, PDF page 9):

What is the best constant for the cyclic-group Bohnenblust–Hille inequality, $\text{BH}_{\Omega_K}^{\leq d}$? As a starting point, is $\text{BH}_{\Omega_K}^{\leq d}$ subexponential in d ?

Their constant concerns functions on Ω_K^N whose Fourier spectrum has total degree at most d .

Answer in later literature

Pellegrino–Raposo answer this affirmatively in arXiv:2608.05366v1. On PDF page 5 they prove Theorem A for the larger class of functions whose Fourier characters involve at most d active coordinates. If $\text{BH}_{d,q}^{\text{int}}$ is its optimal constant, then for every fixed $q \geq 2$,

$$\text{BH}_{d,q}^{\text{int}} \leq \exp \left(c_q \sqrt{d \log d} + O_q \left(\sqrt{\frac{d}{\log d}} \log \log d \right) \right).$$

For a Fourier index $\alpha \in \{0, \dots, q-1\}^N$, its interaction order satisfies

$$|\{j : \alpha_j \neq 0\}| \leq |\alpha|.$$

Thus the total-degree- d class in the source is contained in the interaction-order- d class. Consequently

$$\text{BH}_{d,q}^{\text{deg}} \leq \text{BH}_{d,q}^{\text{int}}.$$

This gives Corollary 1.1 of the later paper, explicitly titled “Answer to the question of Becker et al.”:

$$\text{BH}_{d,q}^{\text{deg}} \leq \exp \left(c_q \sqrt{d \log d} + O_q \left(\sqrt{\frac{d}{\log d}} \log \log d \right) \right) = \exp(o(d)).$$

Under the later paper’s notation, $\text{BH}_{d,q}^{\text{deg}} = \text{BH}_{\Omega_q}^{\leq d}$ from the source. Hence the requested subexponentiality holds for every fixed finite cyclic group, including the intermediate cases $q \geq 3$.

Scope

This is a full affirmative answer to the “starting point” in source Question 2. It does not determine the optimal asymptotic constant, nor does it answer the source’s separate Question 1 about optimal K -dependence of the uniform-norm discretization constant.

References

- [1] L. Becker, O. Klein, J. Sloté, A. Volberg, and H. Zhang, *Dimension-free discretizations of the uniform norm by small product sets*, arXiv:2310.07926.
- [2] D. M. Pellegrino and A. Raposo Jr., *Subexponential Bohnenblust–Hille Inequalities on Finite Cyclic Groups*, arXiv:2608.05366v1 (2026), Theorem A and Corollary 1.1.